

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

NOV-DEC 2012

EC305 Microcontroller & Application

Date: 30.11.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) List applications of microcontroller which are used in real life. [03]
(b) Differentiate low-level language and high-level language. [04]
- Q - 2 (a) Draw pin diagram of 8051 microcontroller. Explain function of following pins. [07]
1. ALE 2. RST 3. EA
(b) Explain memory organization of 8051 microcontroller. [07]

OR

- Q - 2 (a) Explain the operations carried out when the following instructions are executed by 8051. [07]
(i) MOVX @R0,A
(ii) MOVC A, @A + PC
(iii) RLC A
(iv) CJNE A, 50H, L2
(v) JBC 20H, L3
(vi) XCH A, 30H
(vii) XCHD A,@R0, where L2 and L3 are labels.
- (b) State and explain the addressing modes used in each of the following instructions. [04]
(i) MOV A,#25H (ii) MOV A,@Ri (iii) MOV R2,40H (iv) MOVC A,@A+PC
- (c) Explain difference between MOV, MOVX and MOVC instruction with example. [03]
- Q - 3 (a) List SFRs available in 8051 microcontroller. Draw and explain format of TMOD register. [07]
- (b) Explain the significance of the following assembler directives in assembly language programming. [04]
(i) ORG (ii) EQU (iii) DB (iv) END
- (c) Compare CALL-RET and PUSH-POP instructions. [03]

OR

- Q - 3 (a) Distinguish counting and timing requirements. Explain the modes of operation of timer/counter of 8051 with a diagram. [08]
- (b) What is an interrupt? Explain interrupt structure of the 8051 microcontroller. [06]

SECTION - II

- Q - 4 (a) Name any six bit manipulation instructions in Microcontroller 8051. [03]
- (b) List merits and demerits of parallel and serial communication. [04]
- Q - 5 (a) With a neat circuit diagram explain how a 4 x 4 keypad is interfaced with 8051 microcontroller and write C program for keypad scanning. [10]
- (b) Do as direct. [04]
1. DAC 08 output is connected to _____
(i) current to voltage converter circuit (ii) Amplifier
(iii) Attenuator circuit (iv) None of above
 2. ADC 0808 is an _____ bit converter.
(i) 1 (ii) 8 (iii) 9 (iv) 16
 3. _____ Pin of ADC 0808 indicates that the analog signal is converted to digital.
(i) EOC (ii) SC (iii) OE (iv) ALE
 4. 8051 is an example of RISC. True or False.

OR

- Q - 5 (a) Draw the schematic for interfacing a stepper motor with 8051 microcontroller and write 8051 ALP for rotate motor clockwise. [07]
- (b) Write an 8051 program to transfer serially the message "WELCOME TO WORLD OF MICROCONTROLLERS" continuously at a 9600 baud rate. [07]
- Q - 6 (a) Interface 16x2 LCD with 8051. Also write assembly or C program to display "EC Engineering" [07]
- (b) Write an 8051 program to toggle all bits of P0, P1 & P2 continuously with interval of 50 ms using timer 0 in mode 1. (Assume XLAT = 12 MHz) [07]

OR

- Q - 6 (a) Explain self clocking mode of ADC. Interface LM35 with ADC. Write an 8051 program to display temperature on LCD. [10]
- (b) What is the last instruction in most interrupt service routines? What does this instruction do? [04]
